

Boqin Yuan

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Summary

I work on **long-horizon** LLM agents end to end: the **benchmarks** that measure them, the **post-training** that improves them, and the **serving** stack that runs them. I shipped step-level agent evaluation at **Moody's**. I co-authored **SkillsBench** and **Agents' Last Exam**, and contributed to the **Harbor** eval framework. I post-train language and vision-language models with **SFT**, **LoRA**, and **RL**, serve them under production load, and extend open-source **RLHF** pipelines. See more at boqiny.github.io 📄

Education

University of California, San Diego

M.S. in Computer Science

Sep 2025 – Dec 2026

(GPA: 3.96)

University of Illinois at Urbana-Champaign

B.S. in Mathematics & Computer Science; B.S. in Statistics

Aug 2020 – May 2024

(GPA: 3.9)

- Graduate with **Highest Distinction**; James Scholarship & Deans List

Work Experience

Software Engineer Intern (AI Agent)

Moody's Analytics

San Francisco, CA

Jun 2026 – Aug 2026

- Building the **knowledge-iteration** pipeline behind a **Banking Decision Intelligence** agent, a **LangGraph** system that automates analyst workflows, serving in production on **AWS**. Shipped a **FastAPI** + **React** service that scores every **Langfuse** trace with a step-level eval suite, combining rubric **LLM-as-judge** scoring with deterministic tool-call checks to pin each failure on the specific domain knowledge the agent was missing.

Founding Machine Learning Engineer

CambioML (YC S23)

San Jose, CA

May 2024 – July 2025

- Engineered and deployed [Anyparser](#) 📄, a fine-tuned 1B & 2B **vision-language** model for parsing PDFs into structured Markdown (text, tables, charts). Fully fine-tuned and post-aligned with preference data on **8×A100** to improve robustness, beating GPT-4 baselines on **DocVQA** (ANLS). Optimized inference with **SGLang**, serving **150 output tokens/s** on a single **L4** GPU. Deployed as a SaaS on AWS (**ECS** + **Lambda**).
- Core contributor to [Energent.ai](#) 📄, a **computer-use agent (CUA)** sandbox powered by Claude that autonomously executes diverse desktop tasks. Engineered a **multi-agent system** with orchestration, **long-term memory persistence**, and state management, with tool integration and MCP. Leveraged **Kubernetes** to provision isolated per-user sandbox VM sessions, scaling to **1000+** registered users worldwide.

Research Experience

Research Assistant, STABLE Lab

UC San Diego (Prof. Jishen Zhao)

San Diego, CA

Nov 2025 – Present

- **ML systems**: **CodeNib** compiles lexical, dense, and structural repository views per commit, incrementally, giving coding agents bounded context; graph and vector updates run **8.7x/25.4x** faster than rebuild at the median. **Agent memory**: **AMA-Bench** (ICML 2026) measures whether agents carry state across long horizons. **RL training**: **PRO-V-R1** (DAC 2026) post-trains an 8B agent with **SFT+GRPO** on verification rewards, beating GPT-4o on VerilogEval-v2 (57.7% vs 43.0%).

Research Assistant, SSAIL

UIUC (Prof. Minjia Zhang)

Remote

Oct 2024 – Mar 2025

- Extended **DeepSpeed-Chat**, whose RLHF pipeline shipped with **PPO** only, to preference-based post-training. Built a **DPO trainer** scoring chosen/rejected responses against a reference policy with adaptive **KL** control, plus a best-of-N rejection-sampling variant and the data pipeline feeding both. Runs on DeepSpeed's **hybrid engine** and **ZeRO** with **LoRA**.

Selected Publications

[AMA-Bench: Evaluating Long-Horizon Memory for Agentic Applications](#) 📄 ICML 2026

[CodeNib: A Multi-View Data System for Serving Repository Context to Coding Agents](#) 📄 arXiv 2026

[PRO-V-R1: Reasoning Enhanced Programming Agent for RTL Verification](#) 📄 DAC 2026

[Diagnosing Retrieval vs. Utilization Bottlenecks in LLM Agent Memory](#) 📄 ICLR 2026 Workshop on MemAgents

Skills

Languages: Python, C++/C, SQL, JavaScript/TypeScript, Bash

Post-training: PyTorch, DeepSpeed, ZeRO, verl, SFT, LoRA, RL, RLHF, Hugging Face

Agents & Evals: eval harnesses, LLM-as-judge graders, RL environments, MCP, LangGraph, Langfuse, RAG

Serving & Systems: SGLang, vLLM, TensorRT, speculative decoding, quantization, FastAPI, React, Spark

Cloud: AWS, GCP, Azure, Docker, Kubernetes, Terraform, CI/CD, Git, Linux